CAPITEL

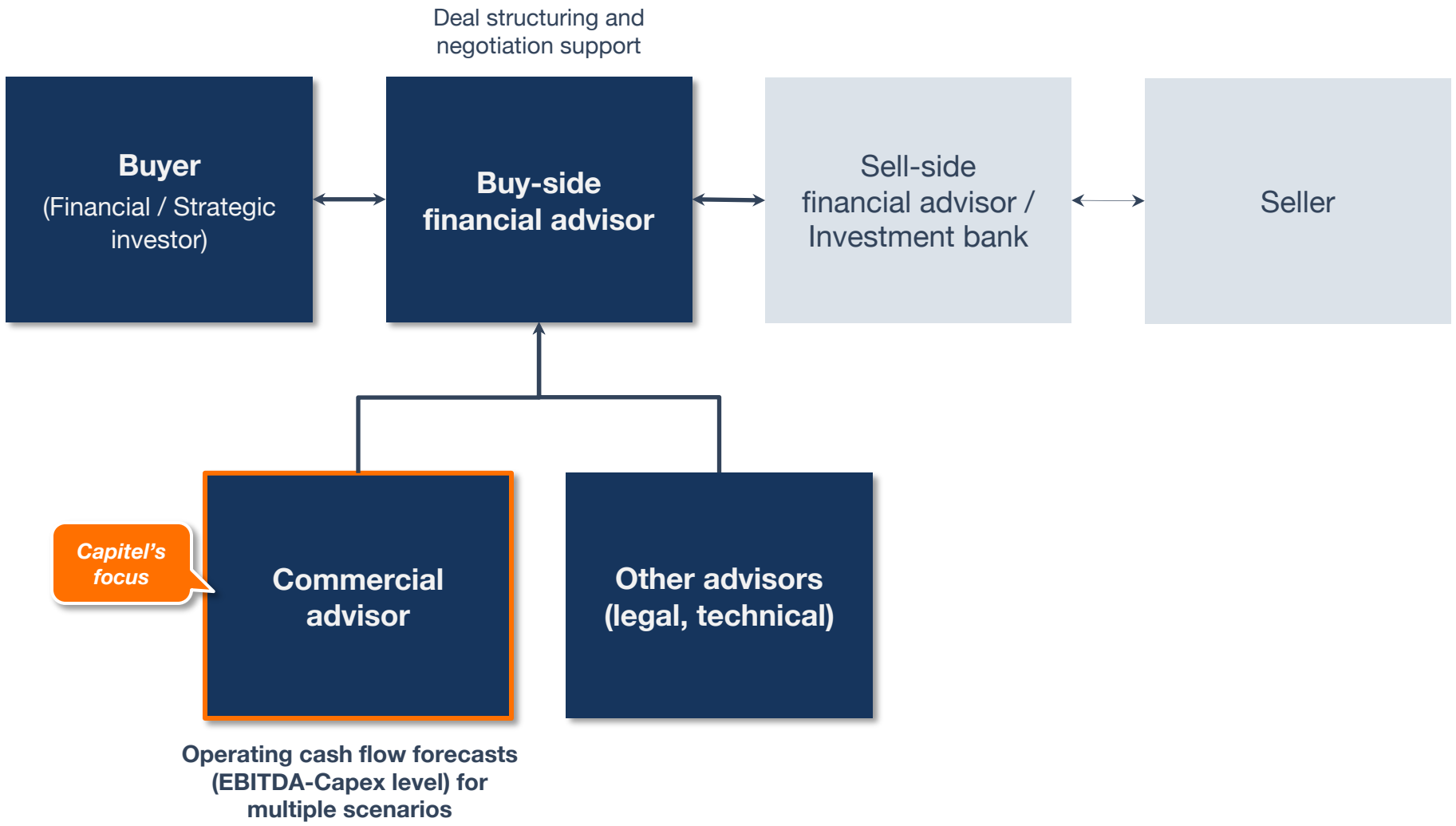
Introduction to Capitel

November 2021

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What we do

We are commercial due diligence advisors for deals greater than $100 million in value, with a focus on buy-side advisory



What we do

More than 80% of our annual engagements are within the Asia-Pacific market

2020-21 Sample Engagements
(not exhaustive)



India
- **Summit Digitel, $500mn bond offering**
- **Brookfield-Digital Realty Trust data center, $1bn+ platform**
- **Brookfield RJio towers, $4bn**
  (largest private equity deal in India)
- **Vodafone fiber network sell side, $2bn+**
- **GPX DC sale to Equinix, sell side**
- **Indus towers**
- **+ more than 15 data center, tower, 5G, cloud engagements**

North Asia
- **KKR-Pinnacle $300mn+**
  (Award for Digital Infrastructure deal of the year in APAC)
- **PhilTower, $100mn+**
- **Globe data centers**
- **Converge home BB**
- **Korea data centers**
- **Taiwan data centers**
- **Rakuten, Japan**

EMEA
- **CellNex towers**
- **Vantage towers**
- **GCC data centers**
- **Madagasacar towers**

Pan-Asia
1. TPG-Towerco

(Indonesia, the Philippines, Malaysia, Myanmar, Bangladesh, Pakistan and Cambodia)

2. PE fund-data center platform

(Indonesia, the Philippines, Malaysia, Thailand and Vietnam)

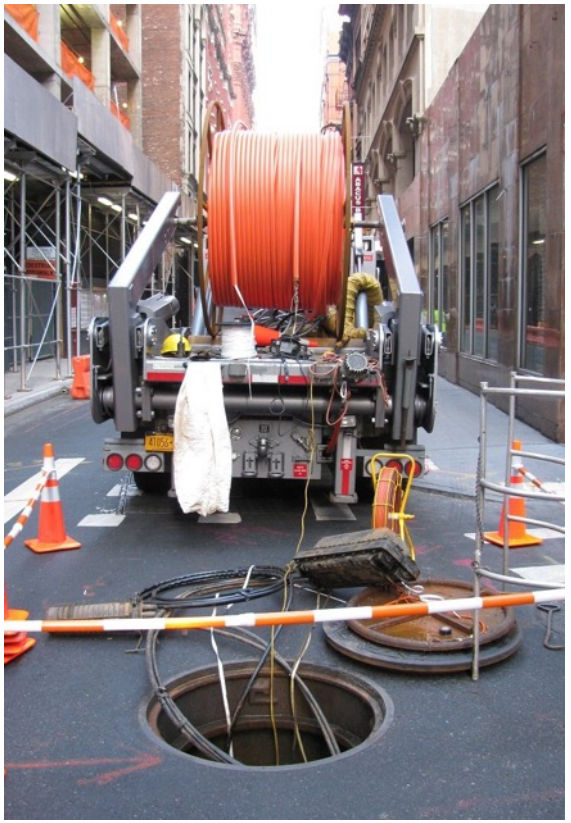
South-East Asia + Australia
- **STP towers, $1bn+**
- **Mitratel IPO, $1.3bn**
- **Data center carve-out, $500mn+**
- **Regional data center platform**
  (Singapore, Indonesia)
- **Starhub-M1 5G**
- **Optus towers**

What we do

We exclusively focus on deals in the digital infrastructure segment



Towers and Street Furniture



Fiber networks and home broadband



Data centers and cloud computing

What we do

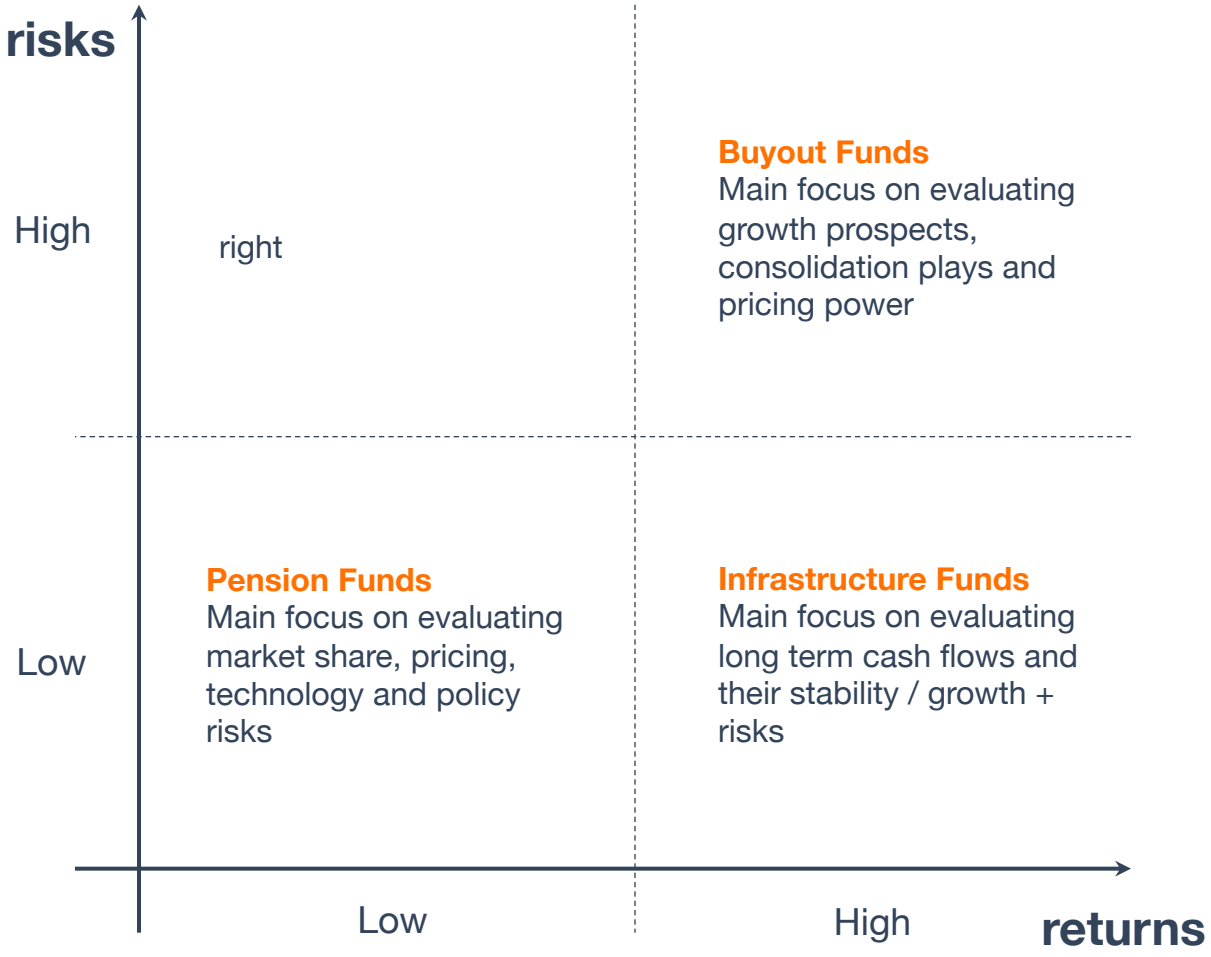
Summary. We specialize in buy side and sell side commercial due diligence of digital infrastructure assets in APAC and EMEA, with a focus on transactions with deal size greater than $100mn

Commercial Due Diligence	Buy-side commercial due diligence Sell-side commercial due diligence Technical due diligence[1]	60+ transactions $100mn to $4bn value per transaction
Asia Pacific	India and South Asia China, Japan, Australia Europe, GCC and Africa	Engagements in more than 15 markets
Digital Infrastructure	Wireless towers Fiber networks Data centers 5G, VRAN	Global PE and Infrastructure funds Major service providers in towers, fiber and data center segments

Note. 1. In partnership with major firms such as Aurecon, Saras, Tilson, Ramboll and others

Sample engagements

We focus only on the most important questions that affect the investment mandate



India I Towers

Canadian Infra fund. Indus Towers, $9bn valuation. Understand the future of Vodafone-Idea and estimate tenancy exits in each of the 700 districts

Commercial due diligence of Indus portfolio of 140,000 towers

Scope

Client: Infrastructure fund

Market: India

Portfolio: Indus towers

Year: 2021

Key Questions

1. **What are the evolution scenarios for Vodafone Idea business in India** – BAU, Regional Player, Acquisition by another operator or Exit? What will be the subscriber base, revenues, OCF and investment capacity of VIL in the BAU and regional operator scenarios?
2. What will be the impact on existing portfolio of Indus in each of the above scenarios? What will happen to the VIL tenancies in case of regional play or acquisition?
3. Assessment of the operational capabilities of the target company and the requirements to roll out a new site in the Philippines in terms of Capex as well as timelines and the potential areas of cost and time over run
4. Other components of the investment case for IC approval including multiple scenarios

Analysis

- District-wise analysis of VIL market positioning, revenues and exit / investment plans
- Mapping of Indus portfolio by districts and estimates of tenancy exits by VIL
- Proximity analysis of Indus portfolio with other towercos to estimate 5G market share
- Analysis of key MSA line items and impact on renewals
- Pricing evolution based on operator financials
- Importance of fiber access and required lease models and organizational capabilities

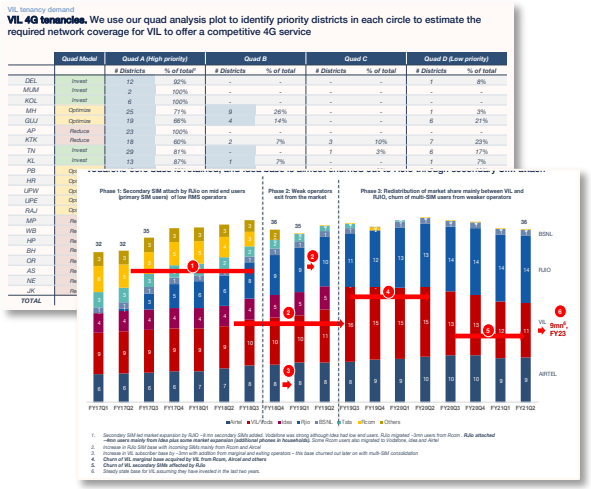
Deliverables

- FY21-30 tenancy forecast by operator and by tower type (GBT, RTT, Street furniture)
- FY21-30 operating cash flow forecast for Indus by VIL expansion and exit scenarios
- All investment case inputs required by the valuation team

Sources

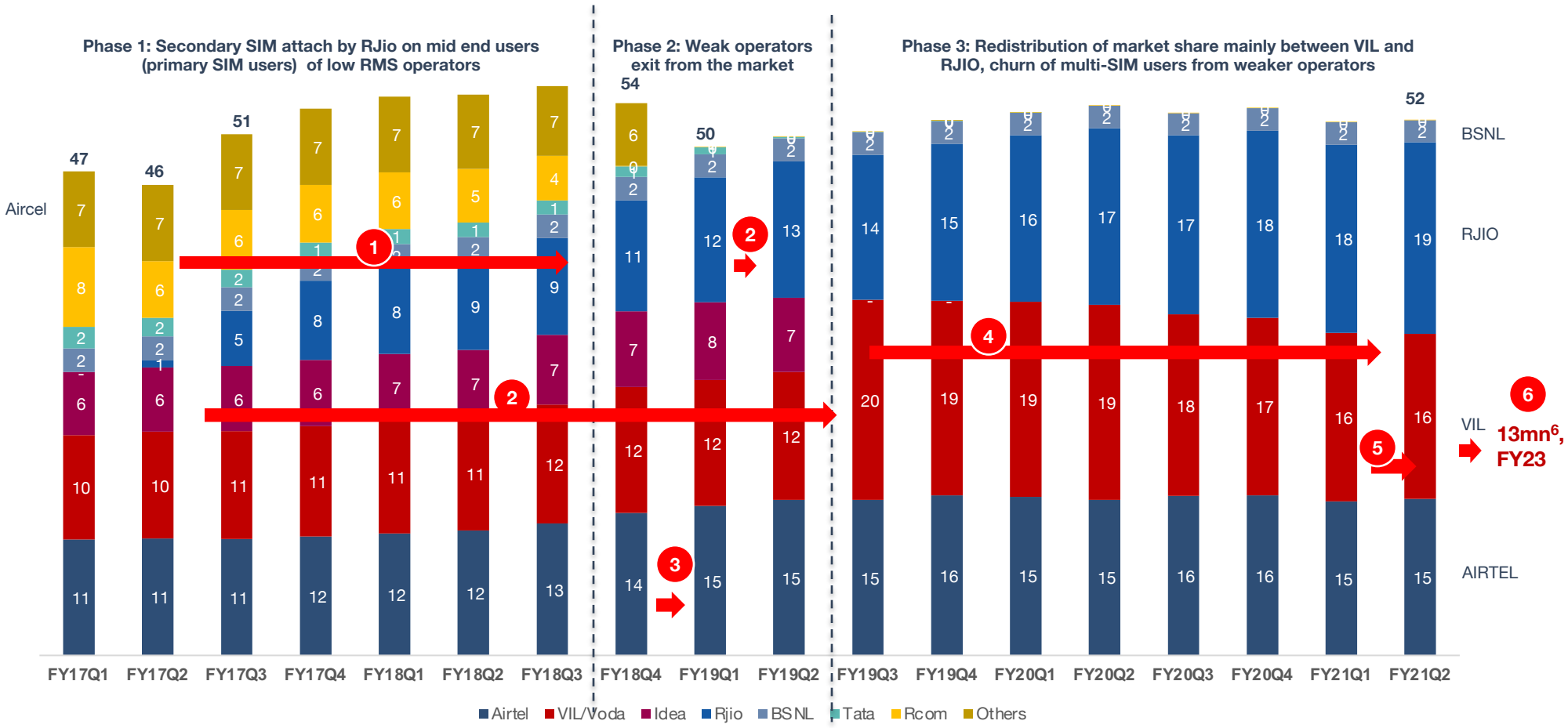
- Prior engagement experience on 20+ tower transactions / investment cases in India
- Lat-Long database of 400,000+ towers
- Industry inputs and operator discussions

Figure. District-wise analysis of Vodafone – Idea subscriber base, share and tenancy exits



Source: Capitel engagement experience

Vodafone. Delhi. RJio attached secondary SIMs to Idea users in the initial stages and VIL also added subs from weaker operators. Churn of weak operator / secondary SIMs will result in 13mn SIM base by FY23



1. *Secondary SIM led market expansion by RJIO ~9 mn secondary SIMs added. Vodafone was strong although Idea had low end users. RJio migrated ~2mn users from Rcom .* ***RJio attached ~4mn users mainly from Idea plus some market expansion (additional phones in households).*** *Some Rcom users also migrated to Vodafone, idea and Airtel*
2. *Increase in RJio SIM base with incoming SIMs mainly from Aircel RJio added ~5 mn SIMs from Aircel and RCom balance. Also additional secondary SIM attach on Idea*
3. *Acquisition of Tata tele by Airtel and addition from exit of marginal operators esp. Aircel*
4. ***Churn of VIL secondary SIM users***
5. ***Market appears to be stabilized with no additional churn expected from RJio except for low end incremental voice users migrating to data***
6. *Steady state base for VIL assuming they have invested in the last two years.*

Source: Capitel analysis

PE fund. Operator DCs. Understand demand available in the local market, and the potential leakage to global locations from sub-sea cables

Data center market entry and investment strategy

Scope

Client: DC platform

Market: The Philippines

Portfolio: Operator DCs

Year: 2020

Key Questions

1. **What is the expected demand by customer type**, namely cloud providers such as AWS, social media providers such as Facebook, Enterprise customers as well as other wholesale / retail users?
2. What will happen to these workloads – will they be served within the market or will be taken out to an aggregation region such as Singapore through sub-sea cables?
3. Given the above context, what is the type of DC facility that will see the maximum utilization?
4. How can the captive data centers of operators fit into the overall mix, and what could be the possible GTM scenarios?
5. What are the key risks to the above thesis?

Analysis

- Demand forecast based on major demand segments at an application level (e.g., Facebook) and allocation of workloads to end user segments such as AWS
- Assessment of sub-se cable capacity and the likely mix of workloads going out of the Philippines – estimates based on latency requirement of workloads and their nature (retail or enterprise)
- High level analysis of operator DC portfolio and any upside from a cloud-connect model

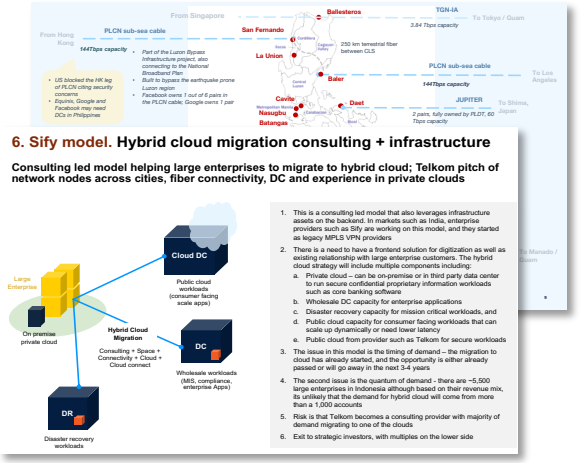
Deliverables

- FY21-30 DC demand forecast by major customers (e.g., Microsoft), enterprises and other B2C providers
- Analysis of regional and global workload hosting model, and estimated distribution
- Analysis of operator DC capacity and revenue upside / expected valuation

Sources

- Prior engagement experience advising funds and DC providers in the region
- Database of sub-sea cables and workload mix for providers such as Facebook
- Database of operator DC capacity
- Industry inputs and operator discussions

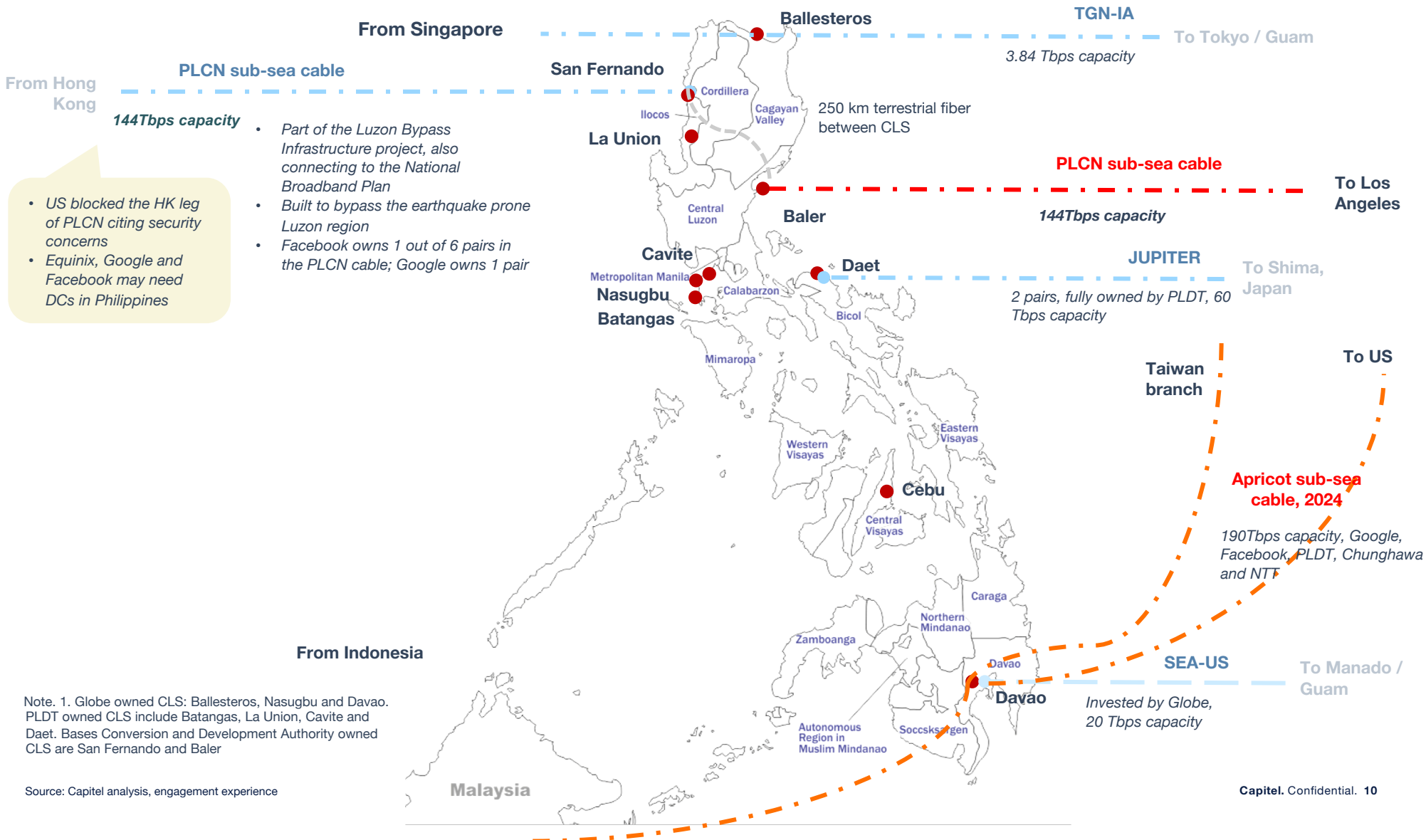
Figure. Analysis of sub-sea cable link capacity by city, and estimated workload migration to global locations



Source: Capitel engagement experience

The Philippines I Data centers

Subsea cables. Facebook's PLCN and Apricot cable and Google's Apricot cables will migrate workloads from the Philippines



- *US blocked the HK leg of PLCN citing security concerns*
- *Equinix, Google and Facebook may need DCs in Philippines*

- *Part of the Luzon Bypass Infrastructure project, also connecting to the National Broadband Plan*
- *Built to bypass the earthquake prone Luzon region*
- *Facebook owns 1 out of 6 pairs in the PLCN cable; Google owns 1 pair*

Note. 1. Globe owned CLS: Ballesteros, Nasugbu and Davao. PLDT owned CLS include Batangas, La Union, Cavite and Daet. Bases Conversion and Development Authority owned CLS are San Fernando and Baler

Source: Capitel analysis, engagement experience

Infrastructure fund. Cloud and Internet. Which cloud provider (such as Google Cloud) will take the majority share of demand from Internet applications and platforms such as Grab and Gojek

Hyperscale DC investment case for micro-markets in India

Scope

Client: Infrastructure fund

Market: Indonesia

Portfolio: Hyperscale data centers

Year: 2021

Key Questions

1. **What is the estimated demand by cloud customers** (AWS, Microsoft, Google, Facebook, SAP, IBM, Oracle and others) within each of the micro-markets such as Jakarta and Batam
2. What is the captive builds by major cloud customers based on industry inputs and announcements
3. What 5 of the open market supply will be good enough to serve demand from cloud customers
4. What will the impact on pricing for hyperscale DCs within each of the micro-markets

Analysis

- Demand modelling using workloads and application forecast by segments such as Social Media, OTT, Internet Platforms, Hybrid Cloud and SaaS
- Estimation of demand by major hyperscale customers based on their network dimensioning frameworks
- Mapping of under construction captive supply by Cloud customers by micro markets to estimate demand-supply gap
- Segmentation of open market supply into Tier 1 and Tier 2 supply to assess the impact on hyperscale pricing

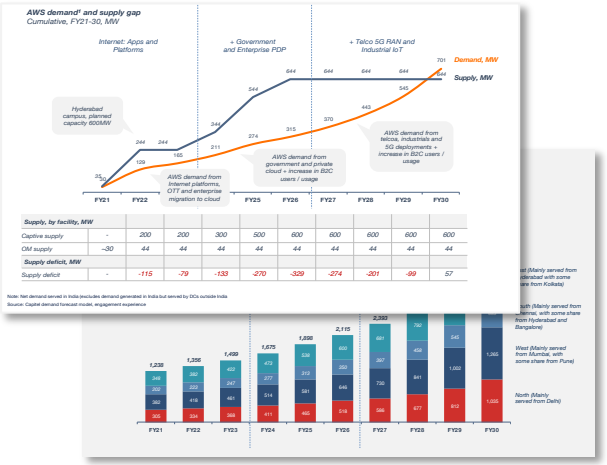
Deliverables

- FY21-30 MW demand forecast by micro markets and by hyperscale customers
- FY21-30 hyperscale pricing ($/KW) evolution by micro-market for Tier 1 and Tier 2 providers

Sources

- Database of all data centers in the major micro-markets in India, including their estimated installed capacity, current construction phase for announced facilities and major hyperscale tenants
- Prior engagement experience advising DC providers and DC investors

Figure. AWS demand within a target micro-market and AWS captive supply, FY21-30



Source: Capitel engagement experience

Indonesia I Cloud

Market share by Cloud Provider. 2025. Major DNBs such as GoJek and Bukalapak are locked by Google and Microsoft; AWS and Chinese clouds are aggressive in consumer and upcoming DNBs

2025, 996 MW	*Social media*	*OTT*	*DNB*	*B2C Ent*	*B2B Ent*	*SMBs*	*5G/Others*
Google	*-*	*-YT + 5%*	***28%*** *(OVO, Traveloka, GoJek), BliBli, Tokopedia)*	*12%*	*12%*	*20% (SaaS products)*	*20%*
AWS	***52%*** *(TikTok, Twitter)*	***60%*** *(Netflix, Iflix)*	***32%*** *(Zalora, OVO, multiple local – Halodoc, GOX)*	*8%*	*9%*	*10%*	*30%*
Microsoft	*10% (LinkedIn)*	*10%*	***16%*** *(Grab, Bukalapak)*	*12%*	*12% (Pertamina)*	*35% (SaaS products)*	*15%*
Facebook	*- (only captive FB apps)*	*-*	*-*	*-*	*-*	*-*	*-*
Chinese + Smaller US clouds	***33%*** *TikTok-Oracle Alibaba-Detikom WeChat-Tencent*	*21% (NBED)*	***14%*** *(Shopee, multiple local)*	***15%*** *(BankMega + other relationship based)*	*10%*	*20%*	*10%*
Local clouds / IT services	*Do not have scale*	*1%*	*2% (smaller Internet cos)*	***10%*** *(OCBC Bank)*	*7%*	*1%*	*5%*
Colo hosting	*5% Does not have flexibility*	*3%*	*8% (non-public cloud workloads)*	***45%*** *(Bank Mandiri, Bank Neo Commerce)*	***50%*** *(Mitsubishi Motors)*	*14%*	*20%*
Total	***279 MW*** *(199 FB + 81 Non-FB[1])*	***72 MW*** *(57 YT + 15 Non-YT[2]*	***321 MW***	***167 MW***	***66 MW***	***46 MW***	***44 MW***

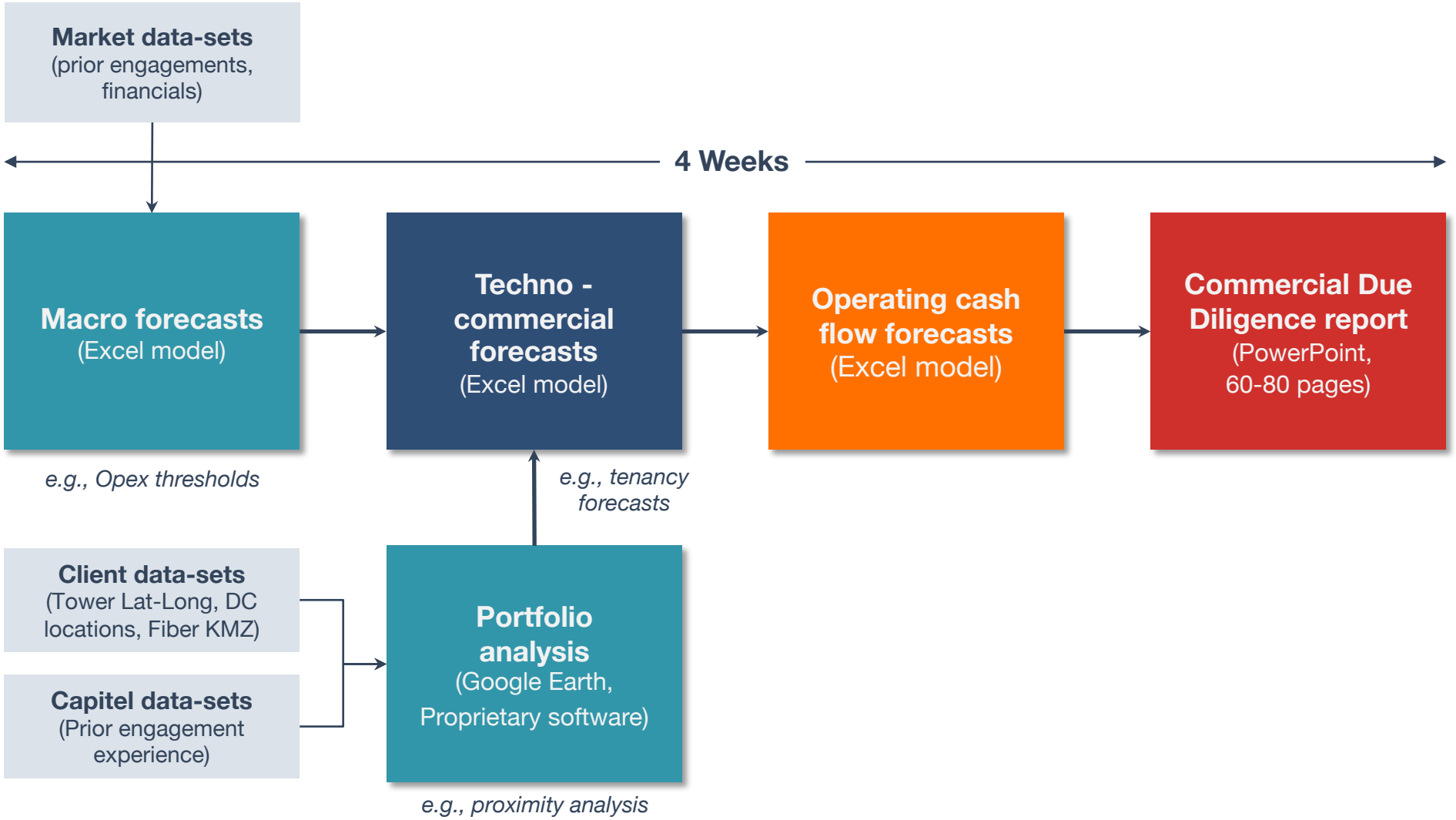
Big platforms are locked by Google and Microsoft (Singapore hosted); AWS is aggregating the long tail of local companies and smaller platforms

Note. CSP shares on non-FB demand from social media 2. CSP market shares on non-YouTube demand

Source: Capitel analysis, engagement experience, company website and reports

Our work

Our engagements are modeling and analysis-led, and typically 3-4 weeks long



Our team

Our teams have clarity of thought and client presence

Key skills

a. ability to think clearly

b. ability to directly deal with senior client teams in India and global markets

c. ability to deal with uncertainty, and

d. comfort with excel modeling

Culture

a. client first mindset

b. zero bureaucracy (or time-sheets)

c. super flexibility (subject to a), and

d. international travel (2+ months / year)

Locations

a. Gurgaon

b. Singapore (if can manage senior international clients)

Levels

a. Analysts (understand industry concepts, basic modeling, attend client calls)

b. Associate (1+ year experience – can handle one part of the engagement such as OCF model and present own part to the client)

c. Consultant (can handle small engagements and manage clients for their own projects)

d. Manager (can handle one complex engagement or multiple small engagements and manage clients for their own projects)

e. Principal (can handle multiple complex engagement and sell new engagements)

f. Partner (find new clients, new markets and new verticals)



Compensation

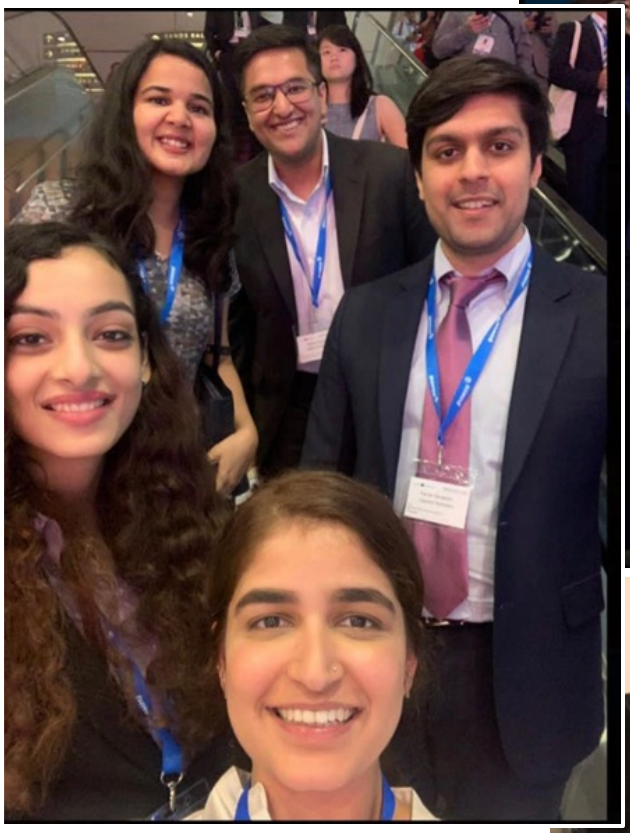
a. CTC INR 15L for Analyst / Associate level

Our team

Culture











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